

Unit 4: Key Features of Linear Functions

1. Does the equation $y + 6 = \frac{5}{4}(x - 4)$ and the function $f(x) = \frac{5}{4}x - 3$ yield the same output?

$$\begin{aligned} y + 6 &= \frac{5}{4}(x - 4) \\ y + 6 &= \frac{5}{4}x - 5 \\ \underline{-6} \quad \quad \underline{-6} \\ y &= \frac{5}{4}x - 11 \end{aligned}$$

No because they have different y-intercepts.

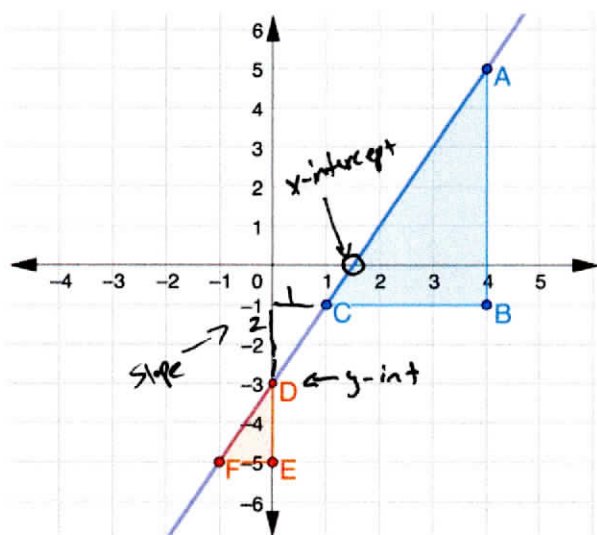
2. Write the equation of the line that is parallel to $y - 11 = -\frac{3}{4}(x + 8)$ and passes through point $(-1, 3)$.

$$\begin{aligned} y - 3 &= -\frac{3}{4}(x - (-1)) \\ y - 3 &= -\frac{3}{4}(x + 1) \\ y - 3 &= -\frac{3}{4}x + \frac{3}{4} \end{aligned}$$

$$\begin{aligned} y - 3 &= -\frac{3}{4}x + \frac{3}{4} \\ \underline{+3} \quad \quad \underline{+3} \\ y &= -\frac{3}{4}x + 3\frac{3}{4} \end{aligned}$$

$$\frac{3}{4} + 3 = 3\frac{3}{4}$$

3. Looking at the graph below identify some key features the line segment $\overline{FA}$



$$y\text{-intercept} = (0, -3)$$

$$\text{Slope} = 2$$

$$x\text{-intercept} = (1.5, 0)$$